

Problem 3

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Problem 1 Assume that $f'(t) = 4t^3 + 2t$ and $f(3) = 5$. Find $f(t)$.

Explanation. This is an Initial Value Problem. First we find the general solution of the differential equation.

$$f(t) = t^4 + t^2 + C.$$

Then, we find the unique solution of the IVP.

$$5 = f(3) = 3^4 + 3^2 + C = 81 + 9 + C = 90 + C$$

$$\implies C = -85$$

and so

$$f(t) = t^4 + t^2 - 85.$$